Joy

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT CODE: 0612 451 07A
CAT 3

29/40

Total marks: 40 **Time: 1 hour 30 minutes**

72%

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Instructions to Candidates

- ✓ Answer ALL questions.
- ✓ Marks are shown in brackets at the end of each question or part.
- ✓ Write clearly and show your working where a calculation is required.
- ✓ Mobile phones and unauthorized materials are not allowed in the examination room.

QUESTIONS

1. Define network testing. **(2 marks)**

2. State two reasons why network testing is important. **(2 marks)**

3. Name the instrument used to measure current without cutting or disconnecting the wire, and state how a voltmeter is connected to the component being tested. **(2 marks)**

4. State the role of Npcap when using Wireshark on a Windows computer. **(2 marks)**

5. Define a VLAN. **(2 marks)**

Question 6 (6 marks)

(a) Define TTL and state what a router does when the TTL of a packet reaches 0. **(2 marks)**

(b) A Windows computer sends a packet with a starting TTL of 128. The packet passes through 9 routers before reaching its destination. Calculate the TTL on arrival and state whether the packet is dropped. **(2 marks)**

(c) Explain how the tracert command uses TTL to discover each router on the path to a destination. **(2 marks)**

Question 7 (6 marks)

A technician runs tracert to a web server and obtains the output below.

```
1    2 ms    1 ms    2 ms  192.168.1.1
2    9 ms    8 ms    9 ms  10.20.0.1
3   26 ms   25 ms   27 ms  41.89.10.5
4  180 ms  178 ms  182 ms  196.201.4.9
5    *       *       *     Request timed out.
6  190 ms  188 ms  191 ms  203.0.113.20
```

(a) Identify the hop where the delay increases sharply and state what this suggests about the network. **(2 marks)**

(b) Give two possible reasons for the asterisks (* * *) shown at hop 5, even though hop 6 replied. **(2 marks)**

(c) Explain why running ping alone would not have shown where the delay occurs. **(2 marks)**

Question 8 (6 marks)

(a) Describe the TCP three-way handshake as seen in Wireshark, stating who sends each packet and the SYN and ACK flag values in each step. **(4 marks)**

(b) Write the Wireshark display filter that would show only:

- (i) the first packet (SYN) of each TCP connection request; **(1 mark)**
- (ii) ICMP Echo Request packets. **(1 mark)**

Question 9 (6 marks)

A technician is commissioning a new office network and records the three findings below.

Finding	**Tool used**	**Observation**
(i)	Voltmeter	A wall socket reads 180 V AC. The expected supply is about 240 V AC.
(ii)	Cable tester	A newly crimped T568B patch cord shows LEDs 1 to 4 and 6 to 8 lighting in order, but LED 5 does not light.
(iii)	Wi-Fi analyser	The signal in the meeting room is measured at -82 dBm.

For each finding, interpret the result and recommend one suitable action. **(2 marks each)**

Question 10 (6 marks)

A school has a Finance department (PC1 and PC2 on ports Fa0/1 and Fa0/2) and a Human Resource department (PC3 and PC4 on ports Fa0/3 and Fa0/4) connected to one Cisco switch. The Finance network is 192.168.10.0/24 and the Human Resource network is 192.168.20.0/24. The school wants the two departments separated.

(a) Write the switch commands to create VLAN 10 named FINANCE and to assign port Fa0/1 to it as an access port. **(3 marks)**

(b) After the VLANs are configured, PC1 (192.168.10.10) cannot ping PC3 (192.168.20.10). Explain why, and describe how communication between the two VLANs can be enabled. **(3 marks)**

END OF QUESTION PAPER

1. Checking network with tools and procedures to confirm its functioning

2 It finds faults before they cause downtime
It confirms performance and security meet requirements.

3 A clamp meter.
A voltmeter is connected in parallel across the component

4 Lets wireshark capture live traffic from the network adapter on windows

What?

5 Divides a switch into separate broadcast domains whatever the physical location of the devices

6(a) an IP header that limits a packet's lifetime in hops. Each router reduces it by 1. At 0, the router discards the packet and send an ICMP message to the sender

(b) 128-9 = 119 - Packet is not dropped.

(c) Sends packet with TTL 1,2,3.

7(a) Hop 4 - Delay jumps from about 26ms to about 180ms. long distance link.

(b) Router at hop 5 may be configured to ignore ICMP.

(c) Ping only gives the total round-trip time to the destination.

8 (a) SYN : client -> server, SYN = 1, ACK = 0
SYN - ACK : server -> client, SYN = 1, ACK = 1
ACK : client -> server, SYN = 0, ACK = 1

(b)(i) tcp.flags.syn == 1 && tcp.flags.ack == 0

(ii) icmp.type == 8

Action?

1

9(ii) 180v is far below 220v - can make equipment reboot

(iii) fibreline 5 has an open circuit - cut and reterminate the connector

(iv) -82dBm is a very weak signal - relocate an access point near the meeting room

2

```
10(a) Switch(config)# vlan 10
      Switch(config-vlan)# name FINANCE
      Switch(config-vlan)# exit
      Switch(config)# Interface fastethernet 0/1
      Switch(config-if)# switchport mode access
      Switch(config-if)# switchport access vlan 10
```

3

(b) The VLANs are separate broadcast domains and a layer 2 switch won't forward traffic between them. Use Inter-VLAN routing.

2

- Layer 3 switch - enable ip routing and create SVIs

Router-on-a-stick - trunk link a router with subinterfaces

1

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